

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI

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Division: C

Batch: 1

Practical No: 1

Roll no: 16

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Subject Name: Deep Learning

Problem Statement:

Install and configure TensorFlow/Keras in Google Colab. Perform data preprocessing, normalization, train-test split, and visualization on a sample dataset.

Algorithm:

Data Preprocessing, Normalization, Train-Test Splitting, and Visualization using MNIST Dataset

1. Start the program.
2. Import the required libraries (TensorFlow/Keras, NumPy, Matplotlib, Scikit-learn, KaggleHub, etc.).
3. Download the Fashion_MNIST dataset using `kagglehub.dataset_download()`.
4. Load the Fashion_MNIST images and labels into memory.
5. Verify the dataset dimensions and inspect the data.
6. Separate the image data (features) and digit labels (targets).
7. Normalize the pixel values by dividing each pixel by 255 so that all values lie in the range 0 to 1.
8. Split the dataset into training and testing sets using `train_test_split()`.
9. Visualize sample handwritten digit images using Matplotlib.
10. Plot the distribution of digit classes to understand the dataset.

11. Display the shapes of the training and testing datasets.
12. End the program.

Code:

```
import kagglehub
import os
import numpy as np
import matplotlib.pyplot as plt

from mlxtend.data import loadlocal_mnist
from sklearn.model_selection import train_test_split
```

```
# Download latest version
path = kagglehub.dataset_download("zalando-research/fash:

print("Path to dataset files:", path)

Using Colab cache for faster access to the 'fashionmnist'
Path to dataset files: /kaggle/input/fashionmnist
```

```
print("Files in dataset:")

for file in os.listdir(path):
    print(file)
```

```
Files in dataset:
t10k-labels-idx1-ubyte
t10k-images-idx3-ubyte
fashion-mnist_test.csv
fashion-mnist_train.csv
train-labels-idx1-ubyte
train-images-idx3-ubyte
```

```
import tensorflow as tf

# Load Fashion-MNIST dataset
(X_train, y_train), (X_test, y_test) = tf.keras.datasets.

# Combine training and testing data
X = np.concatenate((X_train, X_test), axis=0)
y = np.concatenate((y_train, y_test), axis=0)

print("Images Shape:", X.shape)
print("Labels Shape:", y.shape)
```

```
X = X.astype("float32") / 255.0
```

```
print("Minimum Pixel Value:", X.min())  
print("Maximum Pixel Value:", X.max())
```

```
Minimum Pixel Value: 0.0  
Maximum Pixel Value: 1.0
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.2,  
    random_state=42,  
    stratify=y  
)
```

```
print("Training Data:", X_train.shape)  
print("Testing Data :", X_test.shape)
```

```
plt.figure(figsize=(4,4))  
plt.imshow(X_train[0].reshape(28,28), cmap="gray")  
plt.title(f"Digit: {y_train[0]}")  
plt.axis("off")  
plt.show()
```

```
plt.figure(figsize=(8,5))  
  
plt.hist(y_train, bins=np.arange(11)-0.5, edgecolor="black")  
  
plt.xticks(range(10))  
plt.xlabel("Digit")  
plt.ylabel("Frequency")  
plt.title("Fashion-MNIST Class Distribution")  
  
plt.show()
```

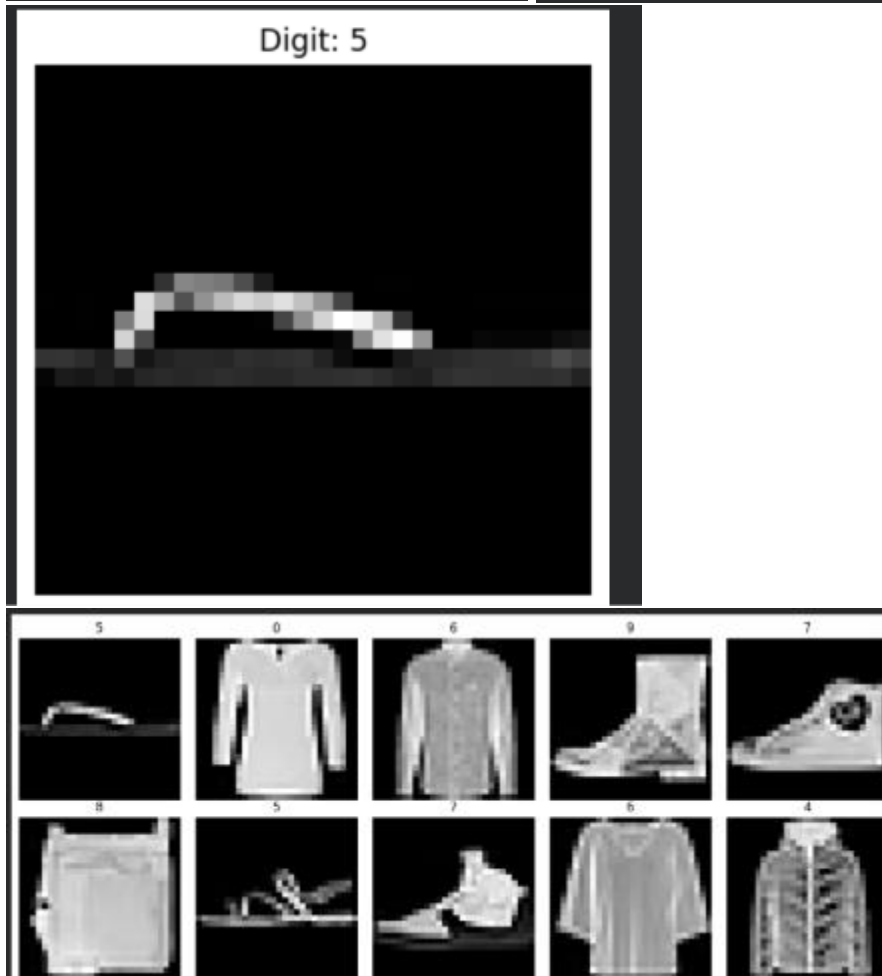
```
print("Training Images :", X_train.shape)  
print("Testing Images  :", X_test.shape)  
  
print("Training Labels :", y_train.shape)  
print("Testing Labels  :", y_test.shape)  
  
print("Pixel Value Range :", X_train.min(), "to", X_train.max())
```

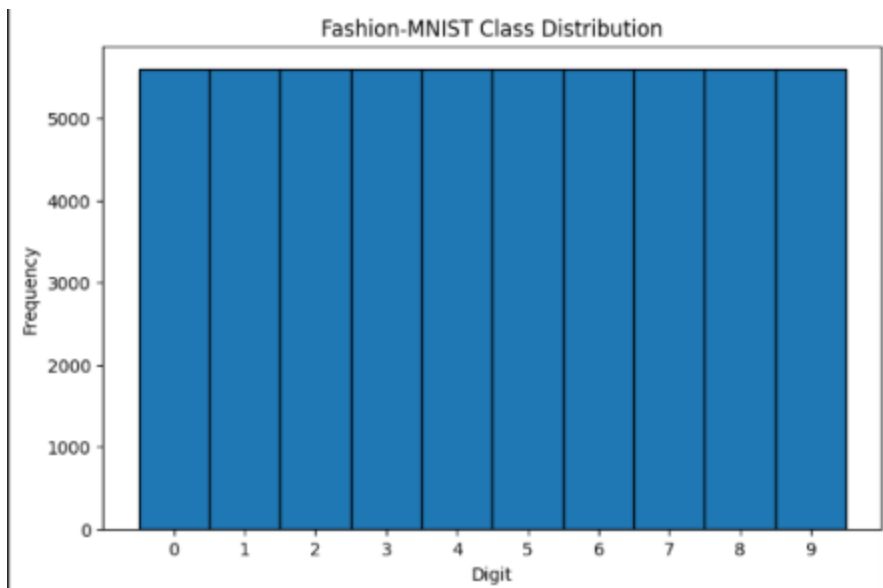
Output :

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-datasets/train-images-1000
29515/29515 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-datasets/train-labels-1000
26421880/26421880 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-datasets/dev-images-1000
5148/5148 ————— 0s 0us/step
Downloading data from https://storage.googleapis.com/tensorflow/tf-datasets/dev-labels-1000
4422102/4422102 ————— 0s 0us/step
Images Shape: (70000, 28, 28)
Labels Shape: (70000,)
```

Minimum Pixel Value: 0.0
Maximum Pixel Value: 1.0

Training Data: (56000, 28, 28)
Testing Data : (14000, 28, 28)





```
Training Images : (56000, 28, 28)
Testing Images  : (14000, 28, 28)
Training Labels : (56000,)
Testing Labels  : (14000,)
Pixel Value Range : 0.0 to 1.0
```